

Malachi Rosario

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EDUCATION

Illinois Institute of Technology

08/2022 – 05/2026

Bachelor of Science Computer Science | Specialization: Data Science

Relevant Classwork: Data Structures and Algorithms | Operating Systems | Database Organization | Systems Programming | Parallel and Distributed Computing | Data Mining | Stochastic Processes | Data Science | Artificial Intelligence | Probability and Statistics | Linear Algebra | Object-Oriented Programming I, II | Graph Theory

Activities: Pi Kappa Phi | SHPE | ACM | Codify | Exelon Summer Institute | CRU

EXPERIENCE

Software Engineer Intern

06/2025 – 08/2025

BlackRock Atlanta, GA

- Developed a full-stack mobile app using Python, FastAPI, and React Native to access and interact with Snowflake-hosted financial data on mobile devices.
- Engineered a dynamic query and filtering caching layer supporting real-time synchronization across arbitrary database schemas.
- Implemented a schema-aware filtering interface supporting text, numeric, and date predicates, enabling flexible querying of Snowflake data from the mobile client.
- Developed stock lifecycle visualization tools to display historical trends and key metrics across investment stages.

Teaching Assistant (TA): CS 116 Data Structures and Algorithms

01/2024 – 05/2024

Illinois Institute of Technology Chicago, IL

- Taught data structures and algorithms in Python to improve time complexity and code optimization, including hash maps, stacks, queues, trees, graphs, and algorithms such as binary search, divide-and-conquer, and greedy.

Full Stack Software Engineer Intern

05/2024 – 08/2024

Treevah Chicago, IL

- Led development of the demo, raising over \$5,000 from investors, including Microsoft, for start-ups.
- Spearheaded the functionality of an online file management system using JavaScript and no-code-based software to build a working prototype.
- Provided the database and infrastructure design for this tech startup for deployment in Microsoft Azure.

Teaching Assistant (TA): CS 116 Object-Oriented Programming II

01/2024 – 05/2024

Illinois Institute of Technology Chicago, IL

- Taught lessons on data structures, algorithms, recursions, object-oriented programming (OOP), and APIs in Java. Guided students, led labs, and offered office hours for Object-Oriented Programming II.

SKILLS

Languages: Python | Java | C++ | SQL | R | JavaScript | TypeScript | HTML | CSS | C | OCAML | Bash

Frameworks/Libraries: React | FastAPI | Streamlit | Pandas | Node.js | Scikit-Learn | TensorFlow | React Native

Developer Tools: Git | GitHub | Azure | Snowflake | Airflow | Docker | Linux | Kafka | MySQL | Tableau | Figma

PROJECTS

Distributed Real-Time Trading System

[View Demo](#)

- Used C, POSIX threads, and TCP sockets to build a distributed real-time trading pipeline across 7 physical nodes on the FABRIC research testbed.
- Built a multithreaded producer-consumer-worker pipeline with mutex-synchronized queuing and dispatch.

S&P 500 Stock ETL Pipeline and Dashboard

[View Project](#)

- Developed an end-to-end ETL pipeline using Python, Airflow, and Azure to collect, transform, and load stock data for S&P 500 companies in real-time. Containerized the system with Docker and provided a Streamlit dashboard.
- Automated extraction from Yahoo Finance, transforming data into structured formats, storing it in Azure Blob Storage and Azure SQL.
- Orchestrated hourly price updates and monthly company refreshes with Airflow.

Real-Time Pitch Predictor Engine

- Built a real-time pitch prediction engine using Python, Kafka, and a trained machine learning XGBoost model.
- Architected a distributed producer/consumer streaming workflow to preserve event ordering.
- Streamed pitch data from MLB StatsAPI through Kafka with one-hot encoding and Joblib for inference.
- Created an interactive Streamlit dashboard to display live pitch predictions and navigate game sequences.